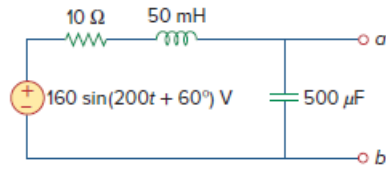


Problem

Obtain the Norton equivalent of the circuit depicted in Fig. 10.106 at terminals a - b .

Figure 10.106:



Step-by-step solution

Step 1 of 9

Refer to Figure 10.106 in the textbook.

From the source voltage, the angular frequency is, $\omega = 200 \text{ rad/s}$.

Convert the source voltage into phasor form.

$$\begin{aligned} 160 \sin(200t + 60^\circ) &= 160 \cos(200t + 60^\circ - 90^\circ) \\ &= 160 \cos(200t - 30^\circ) \\ &= 160 \angle -30^\circ \text{ V} \end{aligned}$$

Step 2 of 9

Determine the impedance of the 50 mH inductor.

$$\begin{aligned} Z_L &= j\omega L \\ &= j(200)(50 \times 10^{-3}) \\ &= j10 \Omega \end{aligned}$$

Determine the impedance of the $500 \mu\text{F}$ capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(200)(500 \times 10^{-6})} \\ &= -j10 \Omega \end{aligned}$$

Step 3 of 9

Draw the circuit in frequency domain.

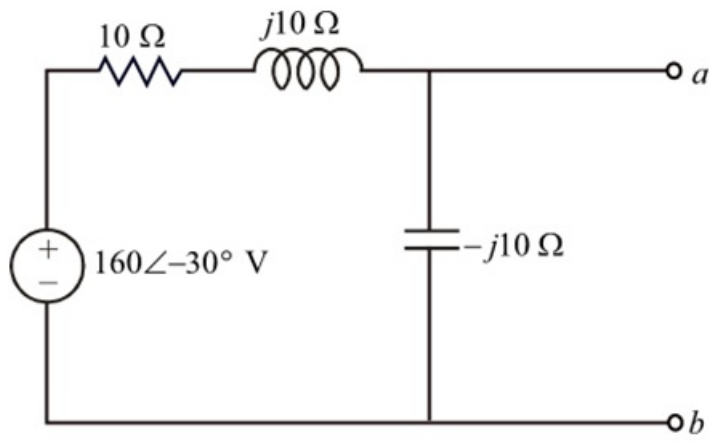


Figure 1

Step 4 of 9

Draw the circuit determine to determine the value of Norton impedance.

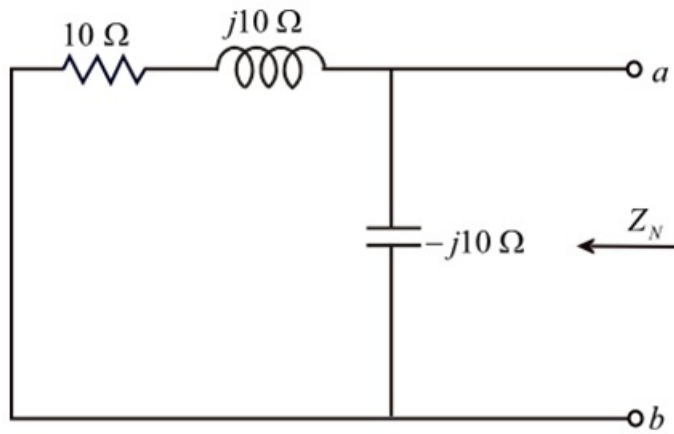


Figure 2

Step 5 of 9

From Figure 2, the value of Norton impedance is,

$$\begin{aligned} Z_N &= (10 + j10) \parallel (-j10) \\ &= \frac{(10 + j10)(-j10)}{10 + j10 - j10} \\ &= 10 - j10 \Omega \end{aligned}$$

Thus, the value of Norton impedance, Z_N is $10 - j10 \Omega$.

Step 6 of 9

Draw the circuit diagram to determine the value of Norton current.

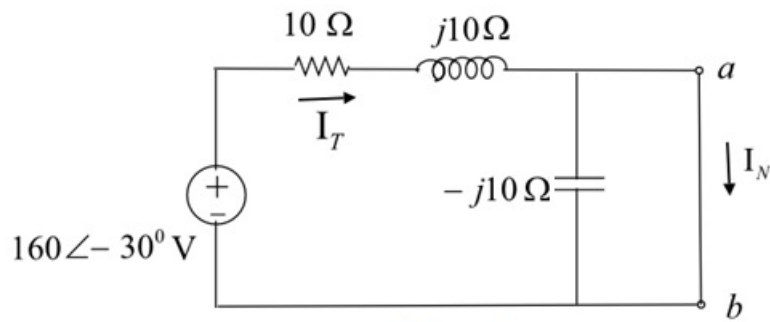


Figure 3

Step 7 of 9

In Figure 3, the capacitor is in parallel with short circuit. The resultant is equals to the the short circuit. Draw the modified circuit diagram.

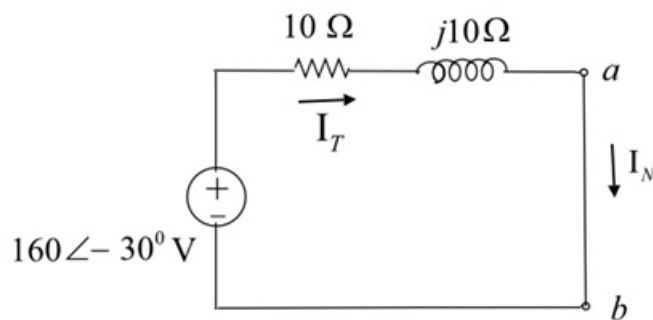


Figure 4

Step 8 of 9

Determine the value of Norton current using Ohm's law.

$$\begin{aligned} I_N &= \frac{160\angle -30^\circ}{10 + j10} \\ &= \frac{160\angle -30^\circ}{14.1421\angle 45^\circ} \\ &= 11.314\angle -75^\circ \text{ A} \end{aligned}$$

Thus, the value of Norton current, I_N is $11.314\angle -75^\circ \text{ A}$.

Step 9 of 9

Draw the Norton equivalent circuit.

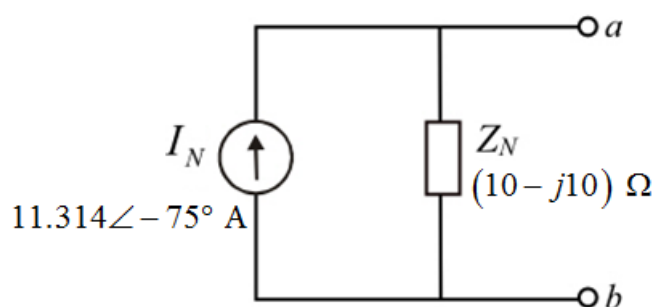


Figure 5

Thus, the Norton equivalent circuit is drawn as shown in Figure 5.